



Fast-lane for planning cycling infrastructure: On the effectiveness and efficiency of cycling infrastructure planning processes

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ABSTRACT

Timely development of cycling infrastructure is essential to achieving societal goals such as decarbonisation and cyclist traffic safety. However, delays in infrastructure project completion persist across many planning contexts, partly due to the infrastructure planning processes. This paper addresses the lack of academic research on infrastructure planning process improvement, specifically for cycling infrastructure, by applying a structured, three-step methodology—process mapping, process analysis and improvement proposal—to the case of Canton Zürich, Switzerland. The paper includes mapping the existing cycling infrastructure planning process, identifying process-related challenges using three decision-making criteria (technical readiness, societal consensus, and political-financial prioritisation), and proposing targeted improvements. Key findings highlight the need for timely planning mandates, early-stage cost overviews, and systematic treatment of uncertainty to enhance planning process efficiency. It is argued that these process modifications can accelerate the realisation of cycling infrastructure projects and improve alignment with long-term strategic goals such as achieving net-zero carbon emissions by 2050. By bridging the gap between planning process design and infrastructure outcomes, this study contributes an approach for analysing and improving planning processes. The findings are relevant for infrastructure planners, policymakers, and researchers seeking to support more effective and efficient cycling infrastructure development.

1. Introduction

Infrastructure planning processes define the way in which societies go about making changes to infrastructure like cycling paths. The processes outline the tasks, which organisation is responsible for each task, each task's inputs and outputs, and the task sequence and dependencies. Infrastructure planning processes are typically defined by legislative bodies entrusted with acting on behalf of society. In alignment with the process, the planning of cycling infrastructure is conducted by planners with the aim to better meet policy objectives, such as decarbonisation, and societal needs, such as improved cyclist traffic safety. Ideally, planning processes facilitate the effective and efficient accommodation of these strategic objectives without any unnecessary delay. To this end, the planning of infrastructure results in infrastructure projects, which are designed, funded, and executed. These projects consist of either construction or modification of infrastructure. However, planned

cycling infrastructure is often completed later than desired.

Against that backdrop, the question can be raised how societies could go about making changes to the planning process that would translate to more effective and efficient changes to infrastructure. However, literature on planning process improvement remains scarce, and literature on improving cycling processes remains even harder to find. To bridge the gap between planning process attributes and project completion, three criteria to funding infrastructure projects are called out: (1) the technical readiness of the project, (2) the level of consensus that the planned project leads to societal needs being accommodated now and in the future, and (3) the financial-political prioritisation allocated to the planned project.

It is challenging to describe multi-stakeholder processes, which involve many planning organisations, let alone make improvements to such poorly documented processes. While these processes may be actively used by planning organisations or exist in their archives, any

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academic discussion on the improvement of planning processes depends on the availability of a holistic process overview. This knowledge transfer between infrastructure planning academics and practitioners is, to date, not taking place in academic journals.

For the first time, this paper reports on an application of a three-step method to proposing improvements for cycling infrastructure planning processes. The method is applied to the cycling infrastructure planning process of Canton Zürich, more specifically to address the challenges it faces with realising projects derived from its 'Cycling Network Plan' in the face of strategic objectives related to a net-zero 2050 and completing their cycling network until 2043. [Section 2](#) includes a review of the current state of the art on process improvements. [Section 3](#) includes a more detailed explanation of the method, including the interviewing, process mapping, and identification of challenges and potential improvements related to facilitating decision-making based on a process resulting in infrastructure projects that are technically ready, have built consensus for the societal value for the projects, and facilitate a political-financial prioritisation of the project. [Section 4](#) includes the steps of the process improvement applied to the case study in Canton Zürich, with explicit links back to the criteria that facilitate decision-making. It finds that a timely mandate for planners, an early-stage network overview of costs, and facilitating early-stage evidence-based decision-making would facilitate more efficient processes. Based on this, the findings are discussed in [Section 5](#) and the paper is wrapped up in [Section 6](#) where the reader is guided through the relevance of the findings and conclusions for academia, industry, and policy-makers alike.

2. A review of efforts to improve planning processes

The aim of this section is to clarify what infrastructure planning and the related processes are, explain the past work done on process improvement, which has primarily focused on improving profit-driven corporate processes, and highlight why improving societally-focused infrastructure planning processes is different and more challenging. As a result, the gap of literature on improving cycling infrastructure planning processes is identified. Furthermore, ways in which process improvements could be assessed are discussed based on criteria that are a prerequisite to a planning process being completed, i.e., conditions related to deciding to finance a planned project.

Infrastructure planning includes all planning activities from identification of societal needs to project delivery. Infrastructure planning processes include all activities to plan the maintenance (e.g., see works from [Chuo et al., 2022](#); [Hässig et al., 2026](#)) and development of infrastructure. In Switzerland, legislative bodies (e.g., the cantonal parliament) task infrastructure planning organisations (e.g., office for mobility) with parts of the development process and the necessary approvals from different stakeholders that must be acquired throughout the process ([Elvarsson and Adey, 2025](#)). Evaluating processes, and consequently improving them, is a challenging task, particularly with a large set of stakeholders with diverging, even conflicting, needs that need to be considered when developing infrastructure.

Improving planning processes requires an overview of the entire process and the different stakeholders involved ([Zairi, 1997](#)). Infrastructure planning organisations, similar to other organisations, strive to improve their processes; infrastructure planning processes, like all business processes, should be effective and efficient to accommodate societal needs ([Adey, 2019](#); [Schraven et al., 2011](#)). A necessary prerequisite to doing so is understanding the planning process, mapping it from end-to-end ([Zairi, 1997](#)).

Process improvements have been widely studied for a corporate setting where there are fewer stakeholders involved in decisions and typically fewer objectives to weigh into the decisions compared to societally important processes (e.g., for planning large infrastructure). Approaches to use statistical methods to improve corporate processes have become increasingly popular. Methods such as Lean Six Sigma aim to use quantitative production data to identify where in the process time

and cost savings can be made in favor of becoming more profitable ([Hahn et al., 1999](#); [Wheat et al., 2001](#)). To expand on this and include more qualitative factors, Cho & Lee propose using a qualitative approach to select the most important factors related to the process and translate them to more quantitative criteria, e.g., delay, cost, or number of failure occurrences ([Cho and Lee, 2011](#)). Other authors argue that qualitative performance indicators can be quantified using "Quality Evaluation Factors" to measure the performance of their processes and strive for their improvement ([Rifkiana et al., 2022](#)). All of these assume that decision-relevant criteria can be reduced to a quantitative value.

Improving infrastructure planning processes, which address societal needs, is not comparable to corporate planning processes, which address the needs of private-market consumers and shareholders. The aforementioned studies describe process improvement for repetitive processes within a profit-oriented business, with a clear beginning and end as well as clear success metrics (e.g., needs of company and customer). However, public infrastructure planning occurs in a more polycentric and political governance environment. The specific result of infrastructure planning processes are projects that consist of either the construction of new infrastructure or modification of existing infrastructure, i.e., interventions ([VSS, 2020](#)). Infrastructure planning processes have unclear beginnings, are iterative and sometimes have no end, and there are unclear, conflicting, and sometimes competing success metrics (i.e., multiple diverse societal stakeholders, e.g., user, service provider, environment) ([Neuman, 2012](#); [Priemus et al., 2001](#)). An infrastructure planning process will identify the societal needs to be accommodated by the infrastructure networks and identify which infrastructure projects are best suited to accommodate these needs in alignment with societal objectives ([Neuman, 2012](#)). To achieve the continual accommodation of the evolving needs of society, planning efforts are made to reconcile the needs of all stakeholders through the different perspectives of different disciplines ([Priemus et al., 2001](#)). Decision-makers, e.g., elected politicians, are responsible for and are judged based on their ability to facilitate the timely delivery of infrastructure to accommodate needs. The provision of infrastructure is, however, based on more factors, such as the interdisciplinary planning outcomes requiring multiple disciplines to align their efforts in accommodating societal needs for an uncertain future ([Akse et al., 2023](#)).

In other fields, complex processes are evaluated based on case studies. Every infrastructure project, and thus every instance of the infrastructure planning processes, is different. Therefore, the retrospective analysis of the process instances largely depends on case studies. For clinical trial processes, which are not often repeated, Grant et al. propose performing process evaluations using case studies, "because it can explore and examine contemporary complex phenomena" ([Grant et al., 2020](#)). Grant et al. list the importance of the interaction between the medical intervention and the context of the intervention as well as highlight the importance of understanding the results based on the trial design.

How can it be argued that improvements have been made? If a process is changed so it becomes more effective or more efficient than before, one may argue that changes have led to improvement. A successful planning process is completed with financing and executing a project. Criteria that lead to the successful financing of a project have been previously discussed ([Elvarsson and Adey, 2025](#)) and are as follows:

(1) The project must be technically ready for implementation, e.g., the project is prepared from beginning to end complying with all requirements such as environmental impact assessments ([Schweizerische Bundesrat, 2018](#)). This means the project is technically ready when it has a building permit or similar, i.e. a formal go-ahead decision from permit-granting bodies. Complying with all requirements and designing the infrastructure according to best practice takes time and benefits from planners having sufficient time and resources to work on the project and make it technically ready. (2) There must be consensus that the project will facilitate the accommodation of societal needs in the uncertain

future, e.g., relevant stakeholders agree that the project is well-suited to accommodate societal needs for times to come. In Switzerland, stakeholder consultations take place throughout the process and explicit consensus is reached if no complaints are filed, or when they have all been settled. This can however be challenging if infrastructure projects are rigidly defined based on constraining assumptions (Cantarelli et al., 2010; Eliasson, 2025). (3) The final criterion that contributes to the financing of the infrastructure project is the political prioritisation to allocate scarce financial resources (Flyvbjerg, 1998, 2007). A project must be prioritised by decision-makers over other projects, like projects in different regions, and other expenditure items, like increased expenditure in education. The criterion is fulfilled if decision-makers are willing to prioritise the project over others and have funding available. If all three criteria are simultaneously fulfilled, projects will get funded on the relevant financing scheme, e.g., a cantonal construction program. These criteria are summarised in Fig. 1.

This list of criteria is aligned with the findings of Sager et al. (2022), who point out that planning infrastructure requires incorporating acceptance of stakeholders and political-financial governance for the technical solution with spatial implications. This list is similar to a list by Velde et al. (2013) including limited budget, higher legal requirements, increased performance requirements, and less public acceptance as pressures on an infrastructure system that complicate infrastructure planning.

3. Methods

To assess a planning process and propose improvements, a three-step method was used. First, the infrastructure planning process was mapped. Second, the decision-making hindrances to realising the planned infrastructure were identified as challenges based on case studies. Third, based on the understanding for the wider context, improvements were proposed that address the challenges and consequently work towards a more effective and efficient planning process. This has been done in other work for road and rail infrastructure development planning processes (Elvarsson and Adey, 2024). Each of these three steps is elaborated in the following paragraphs and summarised in Table 1.

First, the cycling infrastructure planning process was mapped for Canton Zürich. The planning process was mapped across multiple stakeholders, organisations, and different levels of government. This was done because a prerequisite of assessing a process is a holistic

Table 1

An overview of the three steps conducted in this paper.

	1: Mapping	2: Analysing	3: Proposing
Description of purpose	Mapping a holistic end-to-end overview of the process with an insight into the details of each task	Analysing case studies to review past planning decisions and identify challenges related to the planning process	Proposing ways in which the planning process can be improved with a direct linkage to the task that is to be addressed
Description of steps	Review publicly available documents and conduct semi-structured interviews with stakeholders	Identify case studies that were not financed based on the three criteria and understand the underlying reasons	Given the wider context, propose reasonable improvements to the existing process

overview of the process (Zairi, 1997). Furthermore, an insight into the details of each task of the process is necessary, while retaining a higher-level understanding of the purpose of each task. To facilitate such an overview, interviews were conducted with planners involved in the cantonal cycling infrastructure process in addition to a review of publicly available reports and documents. The process was mapped using Business Process Modelling Notation 2.0 (BPMN). The process was modelled based on the publicly available documents published by infrastructure planning organisations, e.g., Canton Zürich's guidelines for developing road projects (Kanton Zürich, 2022b). A list of these documents can be found in the Appendix 7.1. The process was expanded and validated through semi-structured interviews as explained in the Appendix 7.1. In total, five people were interviewed within all of the modelled planning organisations. The interviews were conducted to collect factual information about how processes were set up and to confirm the process drafts by the authors.

Second, the decision-making within the cycling infrastructure planning process is further analysed based on examples of planning process instances, e.g., projects related to the completion of the canton's Cycling Network Plan. This requires a clear link between the process and the resulting decisions related to funding the planned projects. This link is provided by the extent to which the process contributes to the three criteria relevant for infrastructure planning process decision-making, i. e., technical readiness, societal consensus, and political-financial prioritisation. Specifically, it was studied how the planning in these

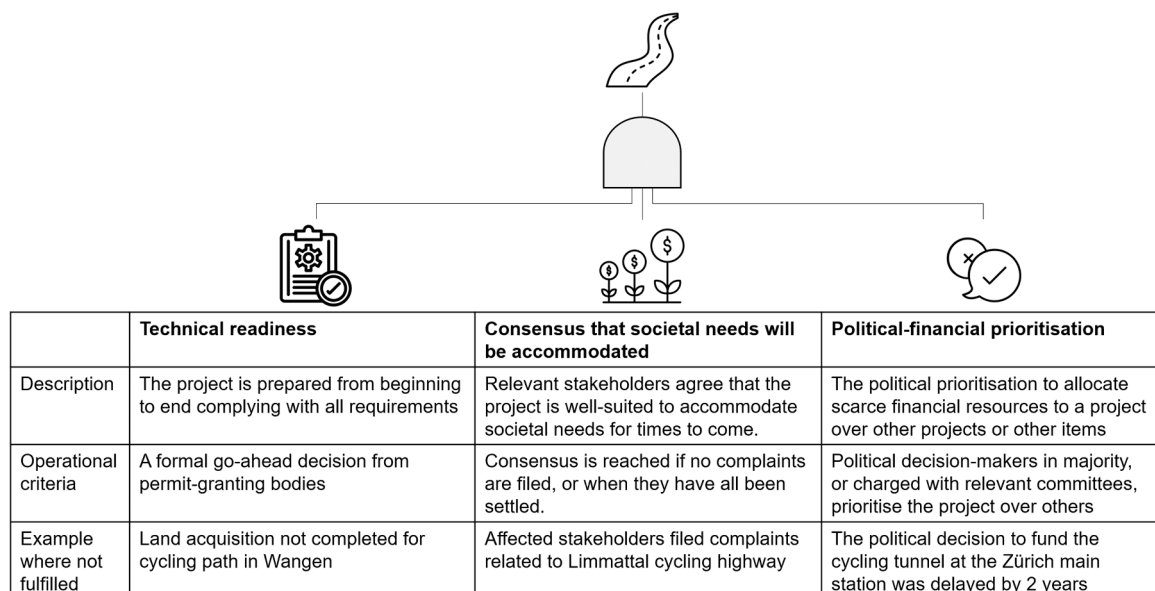


Fig. 1. The three criteria that lead to the successful financing of a project with examples where infrastructure was delayed due to inability to fulfil the criteria.

instances facilitated the three decision-relevant criteria. An output of this was a list of identified challenges to the cycling infrastructure planning process.

Third, based on the identified challenges, improvements for the planning process were proposed. These improvements were linked to the specific tasks that they address and consider the information upstream and downstream in the process (Zairi, 1997). Furthermore, these proposals were related to one or more of the aforementioned decision-making criteria.

The reason why Canton Zürich was chosen for studying is that it has openly available data on their Cycling Network Plan, which made it easier to study. Each canton has the prerogative to set up its own processes in a different way, but each canton's planning organisations may learn from the challenges that Canton Zürich faces in planning its cycling infrastructure. Furthermore, within Canton Zürich, planners may see how planning cycling infrastructure differs from planning other infrastructure, e.g., road and rail.

4. Findings

4.1. General

This section presents the results of the three-step method explained in the previous section. The results of the assessment can support planners and policymakers propose changes towards a more efficient planning process in Canton Zürich.

The planning organisations in Canton Zürich follow a planning process, which, like many other infrastructure planning processes, involves many stakeholders with diverse and competing interests. At the same time, the canton has set clear objectives to better accommodate needs of cyclists and increase the cantonal mode share of trips travelled using bicycles. To achieve this, Canton Zürich has, in alignment with federal law, made a Cycling Network Plan to define an envisioned cycling network end-state that the canton is working towards.

4.2. Mapping the planning process

The cycling infrastructure planning process has been mapped and is visualised in Fig. 2.

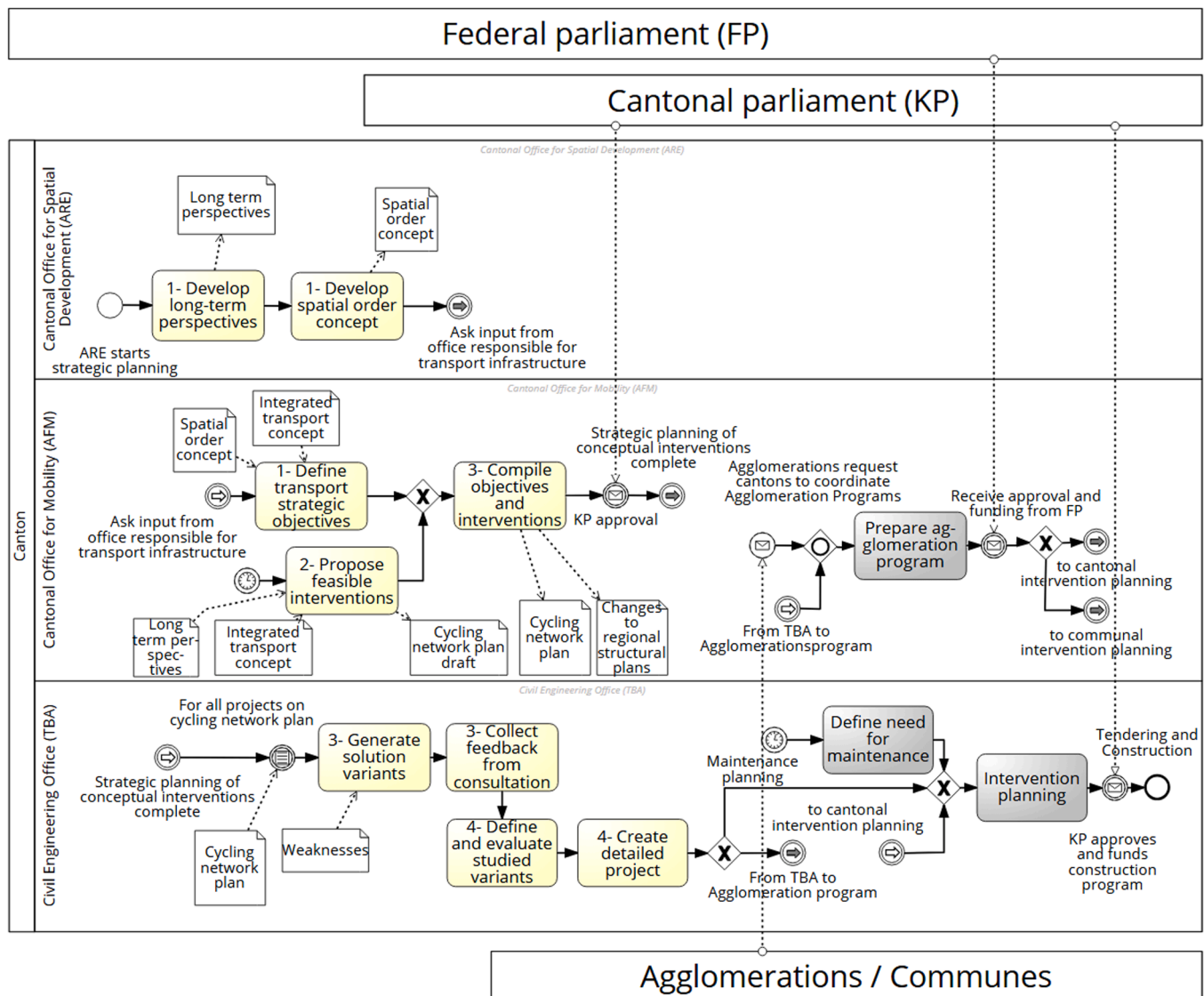


Fig. 2. The cycling infrastructure planning process in Canton Zürich for three different cantonal organisations: the office for spatial development, the office for mobility, and the civil engineering office. The last three tasks are mapped for completeness, but are beyond the scope of this paper.

For infrastructure planning processes, the processes can generally be split into four components, which are also listed in Table 2: 1- define strategic objectives, 2- identify need for change, 3- generate solutions for improvement, and 4- evaluate and prioritise solutions. The table includes a brief explanation of each of the components summarising the tasks and the relevant coordinating stakeholder.

The cycling infrastructure planning process involves multiple organisations responsible for leading different tasks throughout the process. In Canton Zürich, the cycling infrastructure planning processes are carried out by the cantonal office for spatial development, the cantonal office for mobility, and the cantonal civil engineering office. These organisations are the stakeholders who actively shape the planning outcomes and the related projects and are therefore central to the development of the planning process. The legislative parliament bodies on a cantonal and federal level approve project funding and are therefore included in the process diagram. The agglomerations and or communes also play a role in defining process commencement and are key stakeholders in developing projects. Other stakeholders are considered but not explicitly mentioned in the figure. All stakeholders are listed and explained in the Appendix 7.2.

In the part of the process where strategic objectives are defined, the first step is to map the trends and how the canton will develop within the planning horizon (Kanton Zürich, 2024c). Second, the canton will then conceptually order the space available as per its strategic objectives in a Spatial Order Concept (Kanton Zürich, 2023a). Third, the Integrated Transport Concept sets the foundation for achieving a well-functioning transport system that enables sufficient and efficient mobility with a balanced spatial development to minimize any impact on the environment (Kanton Zürich, 2018a). Thus, the Integrated Transport Concept sets focal points where the mobility can be steered and influenced to best achieve those objectives. Based on the Spatial Order Concept and the Integrated Transport Concept, the planners will come up with strategic objectives for all modes.

The strategy for cycling as a mode is defined in the canton's Integrated Mobility Concept, where the cantonal executive council has set a target of 8 % of all trips to be travelled by bicycle in 2030, an increase from 5.5 % in 2015 (Kanton Zürich, 2018a). In 2023, a new federal bill was put in place where cantons are mandated to draft programs for realising connected cycling networks by 2043 (Bundesgesetz Über Velowege, 2022).

When identifying a need for change, the cantonal government will plan projects to better meet the strategic objectives. These projects may

be proposed by either planners or politicians. Against the backdrop of the Long-Term Perspectives and Integrated Mobility Concept, interventions are considered feasible or not (Elvarsson and Adey, 2024). The result of the study is a network plan identifying where cycling paths should be located and where cycling highways should be located.

Based on strategic objectives and the necessary need for change, the cantonal government has defined a Cycling Network Plan. The federal cycling path bill defines the Cycling Network Plan as an envisioned state of a canton's network to be completed by 2043 (Bundesgesetz Über Velowege, 2022). Any difference between this 'should'-state and the current 'is'-state is defined by the canton as a network weakness through a 'weakness study'. All subsequent interventions and projects are generated to alleviate these weaknesses, i.e., work towards completion of the envisioned Cycling Network Plan.

In contrast to rail and road infrastructure projects, cycling infrastructure is defined on Regional Structural plans, created by the regions and coordinated by ARE (Planungs- Und Baugesetz (PBG), 1975). Only cycling highways and cycling infrastructure of national significance are included on the Cantonal Structural Plan. The Regional Structural Plan is subject to consultation feedback from neighbouring regions and cantons where applicable.

For all projects on a Cycling Network Plan, the need for the project has already been established, since the Cycling Network Plan defines the envisioned end-state of the network. Once a planner is mandated to develop a project, the project-level situation is analysed and the goals that will be used to address the need for the project are formulated. Within this task the best variant is identified, based on previous results, and arguments are prepared in favour of the variant. The discussion of the best variant, e.g., in workshops, is important to communicate the results of the process and elicit reactions from stakeholders (Traber, 2015).

Through a public consultation, the cantonal office of civil engineering ensures that all feedback comes across and is addressed to the utmost extent in the final project. Based on all written feedback and any adaptation made to the variants, a project execution mandate is prepared with cost estimates for the entire projecting phase until the end of construction.

Finally, the evaluation is done using a standardised scheme to be comparable across projects over time as described by the canton (Kanton Zürich, 2018b). The scarce resources available to decision-makers are then prioritized accordingly and will result in a development program allocating budget and a timeline for the projects on the program.

After the maintenance managers define the need for maintenance, maintenance projects are prepared and combined with development projects if it is suitable, i.e., they are nearby or provide some other synergies. Lastly, the detailed project is finalised by the civil engineering office, approved with budget by the cantonal parliament, and then sent for tendering and construction.

At the end of the process, cycling projects are to be funded like other infrastructure projects through the 'Construction Program' (Bauprogramm). However, cycling projects are often financed in combination with road projects to make use of synergies. Therefore, the planning process must consider the interface with the planning of other infrastructure types, e.g., road. Additionally, some projects that align with strategic objectives of the federal government may be co-financed by them through instruments such as the agglomeration programs.

4.3. Analyse the challenges related to decision-making

In this sub-section, the cycling infrastructure planning process is analysed to identify decision-making challenges. The analysis is based on past cycling infrastructure projects. These challenges are identified because they impact the effectiveness and efficiency of the planning process.

The challenges are identified based on projects defined in the Cycling Network Plan. Zürich's cantonal Cycling Network Plan defines the

Table 2
The components of the cycling infrastructure planning process in Canton Zurich.

Process component	Description
1- Define strategic objectives	This part of the process aims to make reasonable estimates related to how societal needs could change over time and how such needs are accommodated given possible futures. Coordinator: Cantonal offices for spatial development (ARE)
2- Identify need for change	This part of the process aims to identify where the infrastructure planning organisations may allocate their own scarce resources to best meet the strategic objectives. Coordinator: Cantonal office for spatial development (ARE) and for mobility (AFM)
3- Generate solutions for improvement	This part of the process aims to generate solution variants, estimate the necessary implementation costs, and assess the impact each variant would have on the infrastructure system. Coordinator: Cantonal civil engineering office (TBA)
4- Evaluate and prioritise solutions	This part of the process aims to evaluate the solutions and propose a single project variant so that the project can be appraised and compared to other possible projects on a network level. Coordinator: Cantonal civil engineering office (TBA)

envisioned end-state of the network. The cantonal Cycling Network Plan is the result of the planning process tasks “Propose feasible interventions” and “Compile objectives and interventions” as seen in Fig. 2. The Cycling Network Plan was defined in 2016 (an update since its first version in 2012) and transferred to the Regional Structural Plans. The cantonal Cycling Network Plan is visualised in Fig. 3 based on

openly available data (Kanton Zürich, 2024d). The figure illustrates in a) the entire planned network of 1878 kilometres (km) within the border of the Canton and the hatched area of the cities Zurich and Winterthur. These cities have their own infrastructure planning mandates, hence the network is not shown to be complete there. In b) selected segments of the Cycling Network Plan and municipalities, which are also referred to in

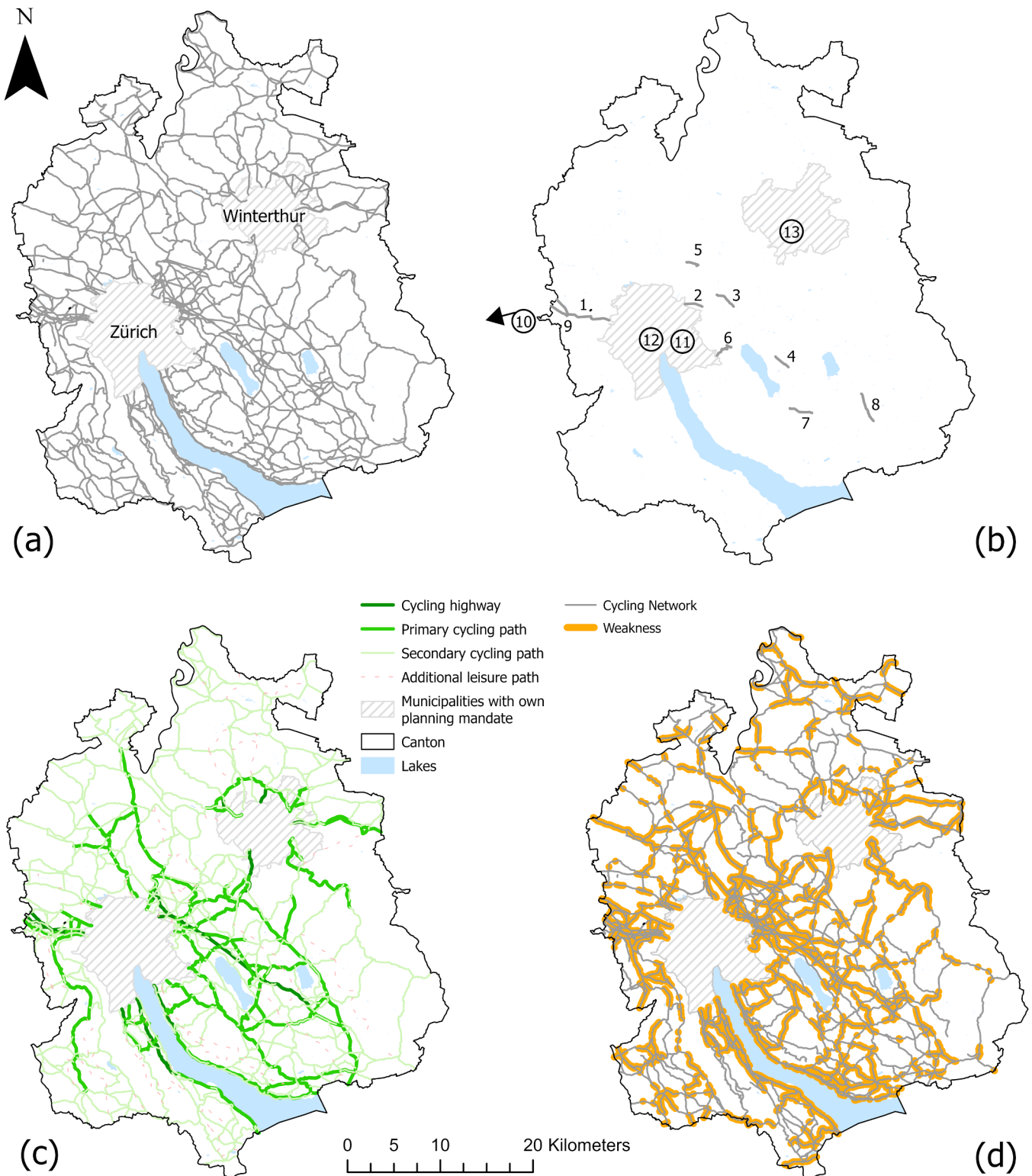


Fig. 3. The cycling infrastructure network decided on by Canton Zurich. The cities of Zurich and Winterthur have retained the planning mandate for infrastructure within their city borders and manage their own cycling networks not shown in this figure. The figure illustrates a) the cantonal network in its entirety, b) location of selected projects referred to in this paper, numbered as per Table 3, c) the different cycling paths by type, and d) a spatial overview of the network weaknesses.

this paper, are visualised. These segments are numbered on the figure, whereby the numbers correspond with the projects listed in Table 3, and are listed and described in more detail in the Appendix 7.3. In c) the current Cycling Network Plan is shown as per the classifications of the cycling network, consisting of cycling highways, primary cycling paths, and secondary cycling paths. In d) the 602 km of network weaknesses are illustrated. These are to be alleviated until 2043 (*Bundesgesetz Über Velowege*, 2022; *Kanton Zürich*, 2024d).

Those segments of the network that need to be improved are defined as ‘weaknesses’ of the network. In order to eliminate the weaknesses of the network, projects are planned and developed that will accommodate societal needs in alignment with the cycling path legislation (*Bundesgesetz Über Velowege*, 2022). To plan the project aiming to eliminate such a weakness requires the budgeting of financial resources to planners within the organisation. Only once the planners are mandated to develop the project can they allocate their time to the planning and project development. Therefore, the technical readiness of the project is directly linked to the time in the process at which planners are provided the mandate to start developing the projects.

There are 602 km which are identified as weaknesses. This does not necessarily imply that the current network consists of 1319 km (1878 km planned network minus 602 km weaknesses) of cycling infrastructure. Instead it means that 602 km of new construction and modification to existing infrastructure is required to alleviate the current weaknesses in the network. For example, a 1 km link envisioned to be a cycling highway, but currently of poorer quality, would be counted as 1 km of weakness. The available data only consists of the georeferenced weaknesses and the aspired network as per the Cycling Network Plan (*Kanton Zürich*, 2024d).

Further spatial analysis was conducted to understand the spatial overlap between the different types of infrastructure owned by different organisations. Consequently, one gets an impression of the planning organisation interdependencies. Using a 10 m buffer around the cycling network links, their proximity to existing roads was measured. The analysis shows that, for example, 417 km of the 602 km are in vicinity of a cantonal road, whereas 74 km require coordination with communal stakeholders. For these infrastructure, the interventions may benefit from synergies if timed with maintenance interventions. At the same time, 111 km are expected to encounter different complications due to them not being in the vicinity of existing road infrastructure, e.g., if they include the development of land with no construction on it, this requires the coordination of multiple land owners. In case of fragmented ownership with diverse interest of the owners, this may slow societal consensus-building. The results of a spatial analysis are summarised in Table 4.

To alleviate the network weaknesses requires financing the interventions. The cycling infrastructure included in a Regional Structural Plan are defined as ‘roads’ by the canton and are therefore funded through the Cantonal Road Fonds (Strassenfonds), unless co-financed by the federal Agglomeration Program (a special national-level financial instrument for infrastructure investment across municipal borders). This

Table 3
The projects numbered in Fig. 3b).

#.	Project	#	Project
1	Zürich Altstetten-Dietikon, Limmattal	8	Wetzikon [Kempton] - Hinwil
2	Wallisellen, Brüttener	9	Dietikon - Kilwangen (Canton Aargau)
3	Dietikon-Wangen, Brüttener	10	Wiler-Utzenstorf (Canton Bern)
4	Neuwiesenstrasse, Uster	11	Mühlebachstr. - Zollikerstr. (Zürich City)
5	Kloten - Bassersdorf	12	Zürich Main station tunnel (Zürich City)
6	Pfaffhausen - Fällanden	13	Töss (Winterthur City)
7	Mönchaltorf - Gossau		

Table 4
Results of analysis of identified cycling network weakness.

Network elements	Length [km]	Share of total
Cantonal cycling network: goal for 2043	1878	
<i>Cantonal cycling network (2024)</i>		
Built	1780	
Built, to be decommissioned until 2043	44	
Unbuilt ^a	98	5 %
<i>Cycling network by type</i>		
Cycling highways	77	4 %
Primary path	365	19 %
Secondary path	1242	66 %
Other leisure path	195	10 %
<i>Weakness</i>		
Total	602	32 %
In vicinity of road ^b	491	
cantonal road	417	
communal road	74	
Not in vicinity of road	111	

^a Unbuilt is the category for links included in the dataset, but labelled ‘planned’ or ‘variant’.

^b ‘Vicinity’ was defined with a 10 m buffer around the cycling path.

is why the cantonal law on roads (*Strassengesetz*) defines the canton’s mandate for the development of cycling infrastructure. In this legislation, a minimum amount for annual expenditure on cycling is set at about 17 million Swiss francs (Mio CHF) and is annually corrected in proportion to the construction cost index (*Strassengesetz (StrG)*, 1981). Members of Zürich’s cantonal parliament wrote a motion (364/2020) for the executive council to pass a bill to increase this amount threefold, first in 2020 and again in 2023, but this was not passed. This highlights a challenge related to the political-financial prioritisation of financial resources to cycling infrastructure.

On average over the last five years, expenditure on completion of the cycling network has been about 20 Mio CHF per year according to cantonal financial reports (*Kanton Zürich*, 2024a). The Canton has published its expenditure intentions for the years 2025–2028, where 20 million CHF are earmarked annually to completing cycling network, including the acquisition of land (*Kanton Zürich*, 2024b). The annual expenditure values reported in Table 5 also include the cost of land acquisition.

For Canton Zürich, the process has been analysed to outline observations that should be taken into consideration when planning. The challenges are explained in detail below, and can be summarised as follows: First, planners lack a clear planning mandate to proactively plan cycling infrastructure, leading to the infrastructure being planned primarily with the maintenance of nearby infrastructure. Second, the Cycling Network Plan does not have a cost overview allocated to all its interventions. Similarly, project costs for some of the projects have only been calculated after the plan was passed (*Herzog*, 2024). Given the current budgeting practices to alleviate Cycling Network Plan weaknesses, this fact that the plans are passed without accurate cost estimates poses the feasibility of the planned intentions into question. Third, it is challenging to build consensus for cycling infrastructure projects as they include addressing divergent stakeholder needs in an efficient way, considering large uncertainties pertaining the future. To analyse these challenges, the text refers to a set of case study projects. An overview of the referenced case studies can be found in the Appendix 7.3.

Challenge 1 – an untimely mandate for planners: Currently, cycling projects are typically developed rather close to the anticipated project

Table 5
Historical cantonal spending on completing cycling network, including the acquisition of land in million Swiss francs per year.

Year	2017	2018	2019	2020	2021	2022	2023
Spending [million CHF]	16.3	13	15.7	15	20.5	19.4	26.5

delivery date. This is different to other transport infrastructure in the canton that take longer (Elvarsson and Adey, 2025). In fact, the initial commitment to build comes later for cycling infrastructure than other infrastructure. Large cantonal road and rail infrastructure projects are typically defined on the canton's Structural Plan, e.g., the A15 Oberlandautobahn highway extension from Uster East to Betzholz. The Structural Plan is a document that is binding for cantons and is approved by the federal government (Planungs- Und Baugesetz (PBG), 1975). Cycling infrastructure is typically included on Regional Structural Plans, which are different to Cantonal Structural Plans. Regional Structural Plans are created by each region within the canton and coordinated and approved by the cantonal ARE. This document is not binding for the cantons. Thus, regional structure plans do not include a commitment for the cycling infrastructure defined there. Meanwhile, the canton is defined as the responsible stakeholder for constructing the infrastructure and, at least partially, maintaining the infrastructure (Strassengesetz (StrG), 1981). Defining location of the infrastructure may be done using different instruments such as the Cantonal Structural Plan, defined by the canton, or the Regional Structural Plans, defined by groups of municipalities and approved by cantons. However, the Regional Structural Plans are approved by the Cantonal Executive Council and not yet discussed by the Cantonal Parliament and are not yet guaranteed financing, as opposed to infrastructure projects on the Cantonal Structural Plan (Planungs- Und Baugesetz (PBG), 1975). While the canton has the responsibility to build cycling infrastructure, there is no instrument that binds them into financing it. This means that financing cycling infrastructure projects is in the hands of the political decision-makers that approve the construction program budget. What follows is that cantonal cycling infrastructure planners often do not get a mandate to start planning cycling infrastructure until later in the process, e.g., when they are asked to plan cycling infrastructure development which would be synchronised with nearby maintenance interventions, e.g., Neuwiesenstrasse in Uster (Stadt Uster, 2025) or Mönchaltorf-Gossau (Amt für Mobilität, 2025). In the absence of a separate planning mandate, economical synergies becomes the dominating factor that determines when and where infrastructure gets built, as opposed to elimination of weaknesses.

The exception to this are selected key projects, e.g., the cycling highways that are planned and strategically publicised to build support for cycling infrastructure and cycling in general, e.g., Limmattal cycling highway (Kanton Zürich, 2018b), cycling highway along the Brüttener tunnel or municipal projects like the Winterthur Töss highway or Zürich's Mühlebachstrasse-Zollikerstrasse highway. However, cycling highways only account for approximately 10 % of the envisioned Cycling Network Plan in Canton Zürich (Kanton Zürich, 2023b). For the other 90 %, it is unclear at what point the planners will be provided with the necessary mandate to start planning, e.g., in the case when the cycling path is along a communal road, a cantonal cycling infrastructure planner may have to wait until the commune decides to maintain the road, at which point the maintenance planner involves the cycling path planner to define requirements for the intervention.

Challenge 2 – a lack of a cost overview:

Cost estimates are important in all stages of the infrastructure planning process. When a planning mandate is granted to planning organisations late in the process, realistic cost estimates are also delivered late in the process. Based on the 602 km of weaknesses to be addressed, the cantonal civil engineering office (TBA) must (re-)design 30 km of cycling infrastructure annually until 2043. About half of all projects have yet to be planned. Constructing 30 km of cycling infrastructure to eliminate weaknesses per year for 20 million CHF would require an average unit cost of no more than approximately 670'000 CHF per km for all interventions. Investing 20 million CHF over 20 years will total 400 million CHF.

Top-down cost estimates are used in early planning stages. However, their usefulness for network cost estimates is limited. Top-down estimates are based on historic data, approximations and rules-of-thumb to

come up with an approximate total estimate of the cost of single and/or multiple projects. In the case of the completion of the Cycling Network Plan, only qualitative project-level assessments have been made public (Herzog, 2024) and quantitative estimates are only publicly available for top-down network-level estimates, which are currently estimated at 800–900 million CHF (Cantonal Executive Council, 2020). If one were to verify these estimates, top-down estimates with mean cost values would not be useful: The cost range for past cycling infrastructure projects studied in this work (see Appendix 7.3) is large, from about 150'000 CHF/km for minor interventions like in Winterthur's Töss project, to 21–23 million CHF/km when larger interventions including engineering structures are required, like in Dietlikon to Wangen or Kloten to Bassersdorf (Regierungsrat Zürich, 2020). Top-down estimates based on past projects are rather approximative. Cost estimates can also change throughout the planning phase for different reasons, e.g., change in material or optimistic evaluations in previous stages. If cost estimates differ largely from the estimates of a previous planning stage, stakeholders may feel uneasy about the project. Any variation in cost estimates between planning stages can affect policymakers' commitment because it will affect the consensus built around the planned project (Rokicki, 2022).

To improve top-down estimates, previous cost data would be useful. However, openly available cost data on previously completed cycling infrastructure is hard to come by. Some data points can be gathered from various sources, e.g., agglomeration programs, press releases, municipality council decisions, public presentations, and the occasional articles in professional journals. At best, total project costs with a general description of what was done can be found, e.g., for a project within the city of Zurich, an existing road segment from Mühlebachstrasse to Zollikerstrasse was converted to a cycling road (where bicycles have right-of-way, yet cars are still permitted). The modification of this cycling road cost 450'000 CHF per km. No land acquisition was necessary in this case.

To better use this historic data for estimating costs of other projects, the projected cycling infrastructure would have to be split into its further components, i.e., a bottom-up cost estimate. The cost components for the projects would be, e.g., unit cost of engineering structure, unit cost of new pavement, unit cost of cycling path on existing infrastructure. The mean total costs of past projects are not an accurate predictor for the total costs of a new project. This is because costs for cycling infrastructure depend on the local context of each project, e.g., if the infrastructure is in an urban or rural setting, the width of the existing cycling path and/or road, if there are spatial conflicts including multiple stakeholders or if there is an engineering structure required. The project costs also depend on the aspired quality of the cycling path: about 10 % of the cycling network is intended to be classified as 'cycling highways', about 20 % is to be 'primary cycling paths', and about 70 % is to be 'secondary cycling paths' (Kanton Zürich, 2023b). While generalising the costs of a typical cross-section (e.g., based on the cantonal design guidelines (Kanton Zürich, 2023b)) of each path type may help in coming up with early-planning stage estimates, verifying that with bottom-up approaches will lead to more accurate cost estimates. However, the effort required for performing bottom-up cost estimates is large, and the data of past projects is scarce, if at all available.

Challenge 3 – Building consensus for an uncertain future: To facilitate a decision in an effective and efficient manner, the building of consensus throughout the process is important. This is very challenging when decision-making builds on previous decisions of rigid nature. Decisions are made throughout the planning process, but previous decisions are typically not open for reconsideration. This is done despite the large uncertainty pertaining to the infrastructure interventions and the consequent accommodation of societal needs, which makes it challenging to convince all necessary stakeholders. This describes the so-called 'lock-in' effect and is particularly relevant for infrastructure with long planning processes, during which societal needs change (Cantarelli et al., 2010; Eliasson, 2025; Elvarsson et al., 2025). For

infrastructure planning processes, the natural response to this is that societies become more averse in making decisions (Eliasson, 2025).

For smaller projects, this lock-in can be mitigated through the use of low-cost pilot projects, as was done in Uster, which transformed a street to a cycling highway in 2024 and made the change permanent in 2025 (Stadt Uster, 2025).

The canton has been transparent in how they assess the societal value of cycling infrastructure and have, for example, openly published their results of the cost-benefit analysis of the Limmattal cycling highway, e.g., communicating cost effectiveness and societal objectives related to the investments, like contributions to net-zero goals (Kanton Zürich, 2018b). However, the lock-in of infrastructure projects results in stakeholder grievances and the accompanying procedures as planning may have become obsolete over time. Examples of this include the further progress of the planning process for the cycling highway in Limmattal, which has been put on hold indefinitely due to the filing of stakeholder grievances to the cantonal administrative court to rule on the project (Freiermuth, 2023). An example of accelerated implementation is the Brüttener cycling highway from Oerlikon to Dietlikon, northeast of the city of Zürich: It is conducted as an auxiliary project to the large Brüttener tunnel rail project, southwest of its southern tunnel portal (Regierungsrat Zürich, 2020). Similar opportunism accelerated the completion of Wiler-Utzenstorf in Canton Bern (Gorregour, 2025).

This way, both the canton and the federal state benefit from project-related cost economies of scale as well as considerable synergies in both parties facilitating the progress of the planning process of each other's infrastructure development. The fact that the cycling infrastructure is in parallel to the railway tracks reduces the number of stakeholders that the canton must interact with and makes the consensus-building easier than otherwise. However, this is an exception to the 'rule' of having multiple stakeholders, each of which should be kept well informed and in agreement with the project objectives at every stage throughout the process. The cantonal constitution ensures the possibility of a referendum to vote on revisiting executive financing decisions for single projects of more than 6 million CHF or any law set.

This would not remove the expected political opposition of allocating funds to the development of one type of infrastructure (e.g., cycling infrastructure) as opposed to other type of infrastructure. In the case of Wetzikon-Kempen a group of political allies attempted to collect signatures to force the hand of regional authorities to remove the cycling path from the Regional Structural Plan (Evangelische Volkspartei et al., 2023).

4.4. Proposing improvements to the planning process

Based on the wider context of the planning process, the following proposals are directed at improving specific parts of the planning process. Built on arguments in favor of better addressing the three criteria related to funding infrastructure projects, the proposals are shown to be reasonable to contribute towards a more effective and efficient planning process. Specifically for the case of cycling infrastructure in Canton Zürich, the legal obligation to construct the network according to the Cycling Network Plan by 2043 does make the matter more urgent.

Proposal 1 – Provide a mandate to planners to technically prepare the projects from early on in the process: Adding the cycling network to binding documents, e.g., the Cantonal Structural Plan, or making the Cycling Network Plan binding would provide the cantonal cycling planners the mandate to plan the cycling network interventions and accordingly ensure the availability of the personnel resources required. This would require a legal change and an adaptation to the planning process.

This would provide planners with the mandate to start the planning work sooner than otherwise so that the infrastructure can be developed in alignment with the strategic objectives in a timely manner. This development is challenging with the large amount of projects in the pipeline. It is even more challenging since the task "Compile objectives

and interventions" leads to the cycling infrastructure being placed on the non-binding Regional Structural Plan and not the binding Cantonal Structural Plan. This is not only important to start planning but many tasks of the planning process depend on this mandate, e.g., to acquire the necessary land.

The 602 km of weaknesses require planning, with stakeholder consultations, before being tendered and constructed. Adequate human resources must be provided to the planning of cycling infrastructure to ensure that the design and planning of the annual 30 km of modification within the network is obtainable within the timeframe. However, the analysis of the process has uncovered a lack of mandate related to improving the network weaknesses identified.

As synergies can be found with pending maintenance activities when developing cycling infrastructure (Zani, 2021), a lot of cycling infrastructure development is bundled with other interventions (e.g., maintenance interventions) in the road space. This situation, where the cycling infrastructure project has not been developed to be technically ready, this would increase the likelihood of planners being rushed in planning the cycling infrastructure in a way that fits the local surroundings on a project level which could result in deviations from best practice, the cycling design standards or inconsistencies in the network (Kanton Zürich, 2023b). While adequate time should ideally be allocated to planners to effectively plan the infrastructure, time is of the essence. The added time required to plan is time that the envisioned unbuilt infrastructure is not meeting stakeholder needs (Elvarsson et al., 2025).

Proposal 2 – Prepare an overview of the expected costs of the anticipated projects at an early stage: This proposes an estimated cost overview of the cycling plan which would in turn ensure policymakers' commitment to the costs from an early stage. While a cost estimate is critical to allocating funds to the project, the costs are currently only estimated late in the process. This means that decision-makers and relevant stakeholders can only be informed about the costs late in the process, risking that there is a change in expectations, potentially affecting the consensus built around the project. To ensure consistent decision-making throughout the organisation and planning process, the decisions would ideally be made in a similar way throughout the planning process (i.e., that decisions in early stages are made in a similar way as in later stages). This would require a similar cost estimation method when assessing the feasibility of the intervention and when prioritising the interventions. An indicative estimation of costs and time plan is useful as a benchmark to guide decision-making downstream in the process (UK HM Treasury, 2022). This will furthermore align all stakeholders to express their commitment the objective of completing the network plan by the given deadline. These objectives should be aligned throughout the organisation (ISO, 2024). Implementing this in the cantonal cycling process could be done in the task 'Propose feasible interventions' in the planning process, where planners are tasked with finding interventions that support completing the Cycling Network Plan.

Without an overview of project costs, projects are prioritised using a qualitative assessment, e.g., using the severity of the weakness or the expected importance of connectivity of the link, which overly focuses on the technical aspects of the network and would fail to address the political-financial prioritisation conducted by decision-makers.

However, collecting this data or computing the costs, which is currently not available, requires resources. With only a scarce amount of financial resources and time available for planning the cycling infrastructure projects, decision makers faced with the challenge of allocating the high investments required for infrastructure are left exposed to the possibility of cost overruns. There is therefore a need for a tool that can efficiently provide an overview of the costs given the local context of each project and, with the more accurate cost estimations, support the prioritisation of the different interventions and decision-making.

One possible way forward would be for policymakers to allocate more resources to the planning organisations, so they can plan the

projects and estimate costs in early stages. If this would lead to accurately estimated costs, this would certainly be effective. It is proposed to supply planners with a tool for more efficient cost estimating by using aerial images to classify the existing infrastructure and understand the state of infrastructure and compare it to the envisioned state of the infrastructure. This approach maps the difference between the current state and end-state considering the local context of the projects allowing for the efficient computationally supported estimation of the costs for each project. This would be particularly suited to making assessments in early stages e.g., identifying that the Pfaffhausen-Fällanden stretch currently has a 7 m wide cross section and requires additional 2 m to meet the aspired safety standards, instead of having to correct cost estimates midway through the process (Amt für Verkehr, 2021b; Kanton Zürich, 2023b). This could then be easily scalable onto the entire network as proposed by Zani and Adey (2025).

Proposal 3 – Implement adaptive planning with explicit consideration of uncertainty throughout the planning process: This requires embracing adaptive planning at all steps of the planning process. Adaptive planning is a long-term planning paradigm addressing the significant uncertainties with which decision makers are confronted (Yap et al., 2025). At the time of funding a project, there must be consensus that the project will facilitate the accommodation of societal needs in the future. Since needs change constantly over time, defining multiple scenarios should be made an integral part of the ‘Long-term perspectives’ developed in the first task in Fig. 2 and be used throughout the planning process. When the planner subjects the planned future to many possible future scenarios, planners cannot devise a single optimal fixed set of actions for the long term, but must remain adaptive, or flexible to change within their strategy to fit the planned entity, e.g., infrastructure, to the new setting. As a consequence, planners may plan a cycling path along a certain route through multiple landowners’ parcels but retain the option of swiftly completing the path along an alternative route if this is not possible in a timely manner. Planners may also seize opportunities, like converting the unused main train station tunnel to a cycling tunnel (Kremer et al., 2025).

Implementing this requires an overview of the entire planning process, because the way that decisions are made, and the factors that feed into those decisions, must be consistent throughout the process (Adey et al., 2025). It therefore follows that scenarios have to be defined early on in the process, e.g., under the task of “developing long-term perspectives” that enable decision-making throughout the process. This facilitates decision-making under uncertainty during the allocation of resources as the “cantonal parliament approves and funds the intervention program”. By facilitating this, adaptive planning tools have the potential to support decision makers in reducing the difference in predicted and current needs at the time of decision, better facilitating the availability of already approved technical solutions and improving the extent to which the planning outcome will be expected to accommodate future societal needs.

Explicitly considering uncertainty makes the appraisal a lot more challenging. This is particularly true in early stages when the uncertainty is large. Furthermore, appraisal in later stages must be consistent with the decision-making in earlier stages. It remains unexplored whether the consideration of uncertainty can positively impact the stakeholders’ willingness to commit to decisions, granted the decisions are flexible based on later development, and consequently the efficiency of planning processes. Furthermore, it remains unstudied what impact it would have on the effectiveness and efficiency of the planning process if planners could plan more adaptively.

5. Discussion

The result of the analysis presented in this paper includes observations related to the ability of and challenges for Canton Zurich’s infrastructure planning process to accommodate the effective and efficient planning of cycling infrastructure. Recommendations are made in this

paper to overcome these challenges, based on analysis of planning process examples in Canton Zurich. These proposals have been linked to the specific changes in the process that would be required and a discussion of the impacts they would have. To underpin the arguments related to the proposals, one could, in future work, make an attempt to simulate the planning process using, for example, discrete event simulations. Such methods would require data related to planning instances to calibrate and validate the models. This could be helpful to assess whether proposed process changes would be useful to facilitate effective and efficient decision-making, and in which cases these changes would be more or less impactful. Discrete event simulations are common when simulating technical processes like construction processes but less explored in those processes that require more elaborate human interactions, like stakeholder deliberations in infrastructure development processes.

While the authors propose looking at case studies through the lens of the different criteria (i.e., technical readiness, societal consensus and political-financial prioritisation) that are required to finance a project, the overfocus on past events may overlook other future challenges. This highlights the importance of having a thorough understanding of the process from end to end, enriched with the wider context of the planning process to provide insights to the challenges and related improvement proposals. Furthermore, the main contribution of this work is related to a demonstration of how processes can be analysed. In order to validate the results themselves, more input would be needed from Canton Zürich. Therefore, although the findings are specific for the case study, the presented methods can be generally used for improving cycling infrastructure processes based on the respective local context. Moreover, the methods can be used for analysing processes and the modification of processes to ensure they can facilitate effective and efficient planning.

Furthering the state of the art on process improvement is particularly relevant for infrastructure planning. Given the persistent gap in literature on methods to identify process improvements for societal processes, the authors propose moving forward by initiating discussions between practitioners and academia. This will support planners and decision-makers in talking about the strengths and shortcomings of their own processes in a structured way. This allows for learning across different planning organisations through future comparative studies, e.g., through a comparison between the process mapped here and the recently published cycling infrastructure planning process of Canton Bern (Canton Bern Tiefbauamt, 2025).

Eliasson (2025) and Cantarelli et al. (2010) call out the role of lock-ins on cost-overruns. The lock-ins occur when a decision has been made in an earlier stage of the planning process that has ‘locked’ the stakeholders in, so they cannot reconsider their previous decision. Since stakeholders do not want to be locked in, it is natural that, all else unchanged, they will defer making decisions as long as possible. If projects do not turn out as they should, the reasonable reaction is to withdraw support for projects vulnerable to uncertainty, all else unchanged. However, if planning processes would embrace adaptive planning diverse stakeholders with divergent needs would be more willing to cooperate within a project while assured that if these projects turn out less favorably, they can be abandoned. However, this is not considered in existing infrastructure planning processes. Infrastructure planning processes can be adapted by society to address these misalignments and enable more effective and efficient planning.

This paper finds that timely mandates for planners, improved cost estimations, and transparent consensus building considering uncertainty can support planners to achieve the following criteria: (1) the technical readiness of a planned project, (2) consensus for the planned project, and (3) facilitate the financial-political prioritisation allocated to the planned project. Better meeting these criteria supports an effective and efficient cycling infrastructure planning process. While the criteria is general to infrastructure planning processes, the findings are specific to the cycling infrastructure planning process in Canton Zurich. ‘Timely mandates for planners’ is specific to when planners are provided the

mandate to develop cycling projects within the canton. ‘Improved cost estimations’ is specific to the fact that Canton Zurich’s Cycling Network Plan seems to have been defined without an overview of the costs needed to eliminate all weaknesses. ‘Consensus building considering uncertainty’ is specific to the fact that the implementation of cycling infrastructure will result in long lasting modifications to the road space in Canton Zurich and having stakeholders onboard with those modifications is necessary. This paper includes proposals to address this based on the specific end-to-end overview of Canton Zurich’s planning process, which may or may not be useful to other planning contexts. These proposals include planning process modifications to better address the aforementioned criteria. However, it may be that these modifications require model-based support that is not available with existing tools, e. g., providing bottom-up cost estimates for entire networks in an early planning stage based on aerial images alone (Zani and Adey, 2025). The exact technical implementation of such a tool is left beyond the scope of this work.

6. Conclusions

This paper contributes 1) the mapping of the cycling infrastructure planning process in the Canton of Zurich, 2) analysis of the process and its ability to facilitate the effective and efficient planning of cycling infrastructure, and 3) suggestions for process improvement. This is the first time that a cycling infrastructure planning process is mapped and its challenges and opportunities for improving it are discussed in a peer-reviewed journal. The proposals to improve Canton Zurich’s cycling infrastructure planning process are formulated based on three general criteria to facilitate effective and efficient infrastructure planning processes. The findings from the case study are not directly generalisable because the propose modifications specifically to Canton Zurich’s cycling infrastructure planning process, but the applied method may be useful for practitioners and academics to make attempts at improving their cycling infrastructure planning processes wherever they are. This paper contributes to and further facilitates knowledge transfer between those interested in gaining overviews to improve planning processes with the goal of meeting societal objectives. This is of value to practitioners hoping to improve their processes and academics wishing to study them.

As a result, this paper reports challenges in the planning process related to, first, planners lacking a clear planning mandate to proactively plan cycling infrastructure, leading to the infrastructure being planned only with the maintenance of nearby infrastructure. Although, in many cases, it will be synergetic to conduct multiple interventions in one project, the time and resources required for strategic development of the infrastructure are, generally, not available if the planning happens so late in the process. Second, the Cycling Network Plan has no cost overview allocated to it and costs for some of the projects have been calculated afterwards, posing the feasibility of their intentions into question given current fixed yearly contribution budgeting approach.

Appendix

A list of documents and overview of interviews conducted

The infrastructure planning process for the canton of Zürich was mapped using information gathered through public documents and interviews with relevant stakeholders. The following Table A6 lists multiple documents used to acquire the necessary information to describe the process. The review was more extensive, the table is not exhaustive, and the documents are not listed in any particular order.

Third, it is challenging to build consensus for cycling infrastructure projects as they include dealing with diverging stakeholder needs in an efficient way to allow grievances to be addressed without delaying the implementation and ultimately achieving policy goals. Furthermore, in future work, efforts should be placed on providing evidence to quantifying the impact of those process improvements, which is left beyond the scope of this work.

Lastly, the study of processes to identify challenges and opportunities for decision-makers has not been widely explored in infrastructure planning. This paper contributes to providing guidance to societies aspiring to improve their legislatively defined processes. The authors encourage process-owners and fellow academics to report on processes to initiate and contribute to discussions on their suitability to achieve societal objectives. This will facilitate further knowledge transfer between practice and academia on process improvement in the interest of achieving societal objectives.

CRediT authorship contribution statement

Adey Bryan T: Writing – review & editing, Funding acquisition.
David Zani: Writing – review & editing, Writing – original draft, Formal analysis, Data curation.
Elvarsson Arnór B: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Arnor Elvarsson reports financial support was provided by Future Cities Laboratory. David Zani reports financial support was provided by ETH Zurich Department of Civil Environmental and Geomatic Engineering. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A6

List of documents used in summarising the process.

Original document title	Author	Description	Citation
Leitfaden für die Kantonale Richtplanung	Federal ARE	A guideline on cantonal structural planning	(Bundesamt für Raumentwicklung (ARE), 1997)
Richtlinien Programm Agglomerationsverkehr (RPAV)	Federal ARE	A guideline on Agglomeration programs	(Bundesamt für Raumentwicklung (ARE), 2020)
Velobahnen: Grundlagendokument	Federal Roads Office	A guideline on definition and conceptual orientation for cycling highways	(ASTRA, 2015)
Handbuch Velobahnen	Federal Roads Office	A handbook on a nationwide design guideline for cycling highways	(ASTRA, 2025)
Raumordnungskonzept	Cantonal ARE		(Kanton Zürich, 2023a)
Langfristige Raumentwicklungsstrategie	Cantonal ARE		(Kanton Zürich, 2024c)
Kantonaler Richtplan	Canton Zürich	The Cantonal Structural Plan for Canton Zürich	(Kanton Zürich, 2022a)
Regionaler Richtplan	Regions of Canton Zürich	The Regional Structural Plan for individual regions of Canton Zürich	(Region Zürcher Oberland, 2022)
Gesamtverkehrskonzept	Office of Mobility	The cantonal Integrated Transport Concept	(Kanton Zürich, 2018a)
Standards Veloverkehr	Office of Mobility	Design guidelines for new cycling infrastructure	(Kanton Zürich, 2023b)
Veloschnellroute Limmattal: Kosten-Nutzen-Analyse	Office of Mobility	A report on a cycling infrastructure cost-benefit appraisal for the cantonal cycling highway project 'Limmattal'	(Kanton Zürich, 2018b)
«Netzhierarchien, Strassenfunktionen und Geschwindigkeiten – Erfahrungen und Herausforderungen »	Office of Mobility	Slide deck from a presentation made by head of Office for mobility on the Cantonal infrastructure projecting process	(Traber, 2015)
Erläuterungsbericht Kantonaler Velonetzplan: Teilprojekt Winterthur und Umgebung	Office of Mobility)	A report on the process for developing the cantonal Cycling Network Plan in the region of Winterthur	(Kanton Zürich, 2014)
Velo- und Businfrastruktur als Begleitprojekte des Brüttener tunnels	Cantonal Executive Council	A presentation on the details of the accompanying infrastructure interventions in connection with the Brüttener rail tunnel	(Regierungsrat Zürich, 2020)

Importantly, interviews with stakeholders helped provide added context to the information acquired. There were three sets of interviews. The first set of semi-structured interviews was in December 2022 and January 2023 with the canton of Zürich and City of Dübendorf and was used to build an initial impression of the infrastructure planning process. A second validation was conducted for more depth on multiple issues related to the infrastructure planning process. This included questions on the specifics of the cycling infrastructure process. A final validation round was held in April and May 2024, where partners were asked to verify the correctness of the process. The partners were asked to be critical, yet lenient in terms of specificity, as the language of the process description is chosen to be general enough for a more general reader and comparability across different stakeholders.

An overview of the interviewed stakeholders is shown in Table A7. Each interviewee was chosen specifically for their function in the infrastructure planning process and their knowledge of the associated tasks. The selection process of interviewees was made to ensure a holistic overview of the planning process for the aspired level of detail.

Table A7

An overview of the stakeholders involved in interviews for information collection.

Stakeholder	Relevant department	Interviewee function
Canton Zürich	Office for spatial development (ARE)	Director of office for spatial development
	Office for Mobility (AFM)	Head of Integrated Mobility
	Civil Engineering Office (TBA)	Project developer
Dübendorf	Transportation planning	Head of City planning
Wetzikon	Transportation planning	Head of Civil Engineering

During the initial set of interviews, questions were asked regarding their understanding of the process, including i) the tasks required and ii) the information and interaction required from other stakeholders within the process. Later, after the process was constructed, the interviewees were asked to estimate the time frames for each task. An overview of when the interviewees were interviewed during the creation of the process is shown in Table A8.

Table A8

Interviewee per interview round. An asterisk indicates that the stakeholder was contacted, but they opted not to respond.

	Round 1	Round 2	Round 3
ARE	X		*
AFM		X	X
TBA		X	*
Dübendorf	X		*
Wetzikon		X	X

The interviews in the first round were informal semi-structured interviews. The guiding questions used differed for each stakeholder depending on their function within the process. There was no standardised way of asking the questions as it depended on the natural flow of conversation during the

interview. The focus on the question related to the description of the process from their point of view, how they viewed other stakeholder inputs, and description of how the coordination and collaboration between stakeholders took place. Examples of these question lists are found in [Table A9](#).

Table A9

Example question lists for semi-structured interviews.

C-ARE	Questions/Topics to be covered during the interview
Introduction	• What is the motivation for planning infrastructure? What are the main points? How binding is it? • Why is there a need for the planning of infrastructure? How does the Canton's process meet that need? • What are hindrances to meeting that need? • How does the Canton see its responsibility in contrast to those of Federal state, Regions and Communes?
Planning in a complex system	• How do you coordinate the needs of the canton and communes? And those of the federal state? • To what extent do you consider transit and road together for your strategic decisions?
Designing the planning process	• How do you identify an inadequate performance of your infrastructure? • If you identify stakeholder needs that are not being accommodated and the ideal solution is not within the cantonal structural plan. How do you proceed? • What are the advantages and disadvantages to your approach? What are the criteria upon which you classify a good planning process?
Responsiveness	• How important is an effective solution? • How important is a quickly implemented solution? • How would you trade these off?
Assessing infrastructure	• For a planning decision, how do you decide which is the best alternative given uncertainty? • Do you set different requirements for cantonal infrastructure (e.g., roads) in rural and in urban environment?
Communes	Questions/Topics to be covered during the interview
Need for change	• how do you determine the need for infrastructure measures (infrastructure expansion/adaptation/deconstruction) for transportation infrastructure? Is your commune also responsible for this needs assessment for cantonal and national roads that run through or touch the city?
Definition of variants	• once the need for the measure has been identified, how is the procedure followed to ensure that users or other stakeholders are satisfied?
Evaluation	• how are the plans evaluated and categorized? Who is ultimately responsible for the decision on the measure for regional infrastructure measures?

Description of stakeholders

Addressing the accommodation of societal needs is the main objective of planners carrying out the infrastructure planning process. The societal needs are based on needs of stakeholders within society. Within society, there are many different stakeholders. These stakeholders are involved in or affected by changes proposed during the planning process. Stakeholders may be categorized into “shaping” stakeholders, “approving” stakeholders, “participating” stakeholders, and “controlling” stakeholders: 1) shaping stakeholders are those who actively shape the planning outcomes and the related projects, e.g., cantonal offices for spatial development, mobility and civil engineering, 2) approving stakeholders are those who approve the projects to achieve the planned outcomes and bear their costs, e.g., the federal or cantonal parliament or the general population through the necessary popular approval processes, 3) participating stakeholders are those who participate in the planning tasks to develop projects, e.g., public transport service operators, rail infrastructure operators and road operators, 4) controlling stakeholders are those who control whether the planning process and outcome meets their needs and requirements, e.g., rail regulator, residents, nature preservation groups and others. All stakeholders considered in the planning process are summarized in tabular form in [Tables A10–A13](#).

Table A10

Description of shaping stakeholders involved in the cycling infrastructure planning process.

Stakeholder	Stakeholder description with example related to Canton Zürich, when applicable
Cantonal infrastructure planning organization	The canton is responsible for the following tasks: (A) the infrastructure that it owns and operates and (B) coordinating transportation operators in its regions. The canton also (C) represents the interests of all stakeholders, sheds light on and mediates the views of different stakeholders to find a good solution for a region. In Canton Zürich, an office for spatial development defines the strategic direction of the spatial order of the space available to the canton. An office for mobility is responsible for the integrated planning to separate the system-oriented development planning from the conventionally asset-oriented construction and maintenance. The canton is generally responsible for representing communes, regions, and agglomerations in the process of developing agglomeration programs, funded by federal government. The cantonal Civil Engineering Office (Tiefbauamt) is involved in the process by developing the projects defined in the planning documents.
Communal planning organization	The communes plan own transportation infrastructure but have limited influence on the regional development by the canton. When communal infrastructure is of regional importance, agglomerations can submit proposals for agglomeration programs to the federal office for spatial development for approval and financing. Otherwise, communal stakeholders are generally represented at the Cantonal Civil Engineering Office.

Table A11

Description of approving stakeholders involved in the cycling infrastructure planning process.

Stakeholder	Stakeholder description with example related to Canton Zürich, when applicable
Federal Parliament	The federal government prioritizes some cycling infrastructure projects through the national road and agglomeration funds (NAF). These projects are developed by the canton, region, and communes and approved as part of agglomeration programs.
Cantonal parliament	Cycling infrastructure are defined as roads by the Roads Law (Strassengesetz) making them assets of the canton, and they are in turn responsible for their construction and maintenance. Cantonal authorities become involved in communal and national planning efforts and may participate in costs of the infrastructure based on constraints set, on a case-by-case basis.
Communal parliament	Communes form the regions. The regions define the regional structural plans. Accordingly, the communal governments have a say in the design of the infrastructure network and proportionally participate in the costs.

(continued on next page)

Table A11 (continued)

Stakeholder	Stakeholder description with example related to Canton Zürich, when applicable
General population	Projects must be placed under public scrutiny through formal approval processes. If any of the affected stakeholders feel their needs are not adequately met throughout the planning process, they can direct their grievances to the Canton during the Approval Process or ultimately the Cantonal Administrative Court. In the interest of earning popular approval, planning organizations often encourage active participation of the local population to elicit and address their stakeholder needs.

Table A12

Description of participating stakeholders involved in the cycling infrastructure planning process.

Stakeholder	Stakeholder description with example related to Canton Zürich, when applicable
Road operators	The communal civil engineering offices are typically responsible, or contracted, for the operations of the road. They participate in planning process through clear operative requirements and ability to accommodate future societal needs.

Table A13

Description of controlling stakeholders involved in the cycling infrastructure planning process.

Stakeholder	Stakeholder description with example related to Canton Zürich, when applicable
Planning regulators	The Cantonal Office for Spatial Development (ARE) ensures that strategic objectives are met. Cantonal regional plans are coordinated by the ARE. Earlier in the process, the ARE coordinates the setting of strategic objectives, controlling and ensuring that strategic objectives are met. Similarly, the Cantonal Office for Spatial Development (C-ARE) controls that communes meet strategic objectives. The office for mobility is also tasked to strategically balance the investments and incentives between different modes according to cantonal strategic objectives.
Users	Participate in public processes during the planning process and voice their (dis)satisfaction during operation.
Residents	Participate in public processes during the planning process and voice their (dis)satisfaction during operation.
Transportation policy groups	Participate in public processes during the planning process and voice their (dis)satisfaction during operation. e.g., ProVelo (www.pro-velo.ch),
Nature preservation groups	Participate in public processes during the planning process and voice their (dis)satisfaction during operation.

A list of case studies considered

In this paper, the following case study projects have been considered and mentioned explicitly in the text. The numbering in [Table A14](#) is in alignment with [Table 3](#) and [Fig. 3b](#)). The unit cost of the projects should be read with caution as unit costs always depend on local context of each project.

Table A14

List of projects referred to in this paper, their estimated cost and length.

#	Project ¹	Estimated cost [Million CHF]	Length [km]	Unit cost ² [MCHF/km]	Phase ³	Description	Source
1	Zürich Altstetten-Dietikon, Limmattal	50	12	4.2	P	A cycling highway planned for construction in 2024. Delayed due to complaints filed by affected population.	(Kanton Zürich, 2018b)
2	Wallisellen, Brüttener	23.2	2.7	8.6	P	A cycling highway planned and to be executed in collaboration with SBB	(Regierungsrat Zürich, 2020)
3	Dietlikon-Wangen, Brüttener	48.5	2.1	23.1	P	A secondary cycling path with planned start in 2032. Includes an underpass.	(Regierungsrat Zürich, 2020)
4	Neuwiesen-strasse, Uster	1.2	0.4	3.0	P	Conversion of a street into a road dedicated to cycling along a cycling highway. Pilot project run 2024, implemented 2025.	(Stadt Uster, 2025)
5	Kloten - Bassersdorf	67	3.2	20.9	P	A primary cycling path planned from airport to neighboring commune via densely built areas and engineering structures.	(Amt für Verkehr, 2021a)
6	Pfaffhausen - Fällanden	7.5	1.1	6.8	P	A cycling path requiring engineering structures to accommodate for altitude climb.	(Amt für Verkehr, 2021b)
7	Mönchaltorf - Gossau	5	2.5	2.0	P	A new cycling path to close gap in network plan.	(Amt für Mobilität, 2025)
8	Wetzikon [Kempton] - Hinwil	3.3	1.7	1.9	C	Improvement of existing cycling infrastructure to alleviate network weakness.	(Amt für Mobilität, 2024)
9	Dietikon - Kilwangen (Canton Aargau)	8.4	3	2.8	C	Continuation of the Limmattal cycling route from Zürich City westwards across cantonal borders. Start 2024–2025.	(Amt für Mobilität, 2023)
10	Wiler-Utzenstorf (Canton Bern)	1.4	1	1.4	O	Construction in 2024. Expedited timeline as synergies could be found with rail infrastructure project.	(Gorrenbourg, 2025)
11	Mühlebachstr. - Zollikerstr. (Zürich City)	1.13	2.5	0.45	O	Conversion of a street into a road prioritising cycling.	(Stadt Zürich, 2024)
12	Zürich Main station tunnel (Zürich City)	40	0.3	133	O	An unused tunnel converted to cycling use with parking garage for bicycles.	(Kremer et al., 2025)
13	Töss (Winterthur)	0.4	2.6	0.15	O	Minor interventions undertaken to allocate road space to cycling	(Elsener und Hug, 2024)

¹ Projects number 9–13 are located in part or entirely outside of Canton Zürich's planning mandate.

³ For the column 'Phase', the letter 'P' stands for 'Planning phase', 'C' stands for 'Construction phase' and 'O' stands for 'Operation phase'.

Data availability

Only public, open data was used and has been cited accordingly in the paper.

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